Denominator Audit of Bell/CHSH Statistics: Protocol Sensitivity to Pairing-Rule Definitions

贝尔不等式的分母审计：配对规则定义的协议敏感性分析

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Source code & data / 源代码与数据： https://github.com/tomnattle/chain-explosion-model
Clone this repository and run the commands in §6 from the **repository root**. / 请克隆该仓库，在**仓库根目录**执行第 6 节命令。

ENGLISH

Bell inequality violations are historically cited as proof of non-locality. This paper presents a "Denominator Audit" of Bell/CHSH statistics using NIST experimental data. We show that the reported value is sensitive to coincidence-pairing protocol definitions and denominator bookkeeping choices under a shared event stream. On the published CSV grid, changing only the pairing tolerance in **NIST grid-index units** (0.0 vs 15.0) shifts the binary CHSH statistic by ~+0.50 while holding the event list fixed.

中文对照

贝尔不等式的违背在历史上被视为非局域性的证据。本文利用 NIST 实验数据对 Bell/CHSH 统计进行了“分母审计”。我们显示：在同一事件流上，报告值对符合配对协议定义与分母记账选择具有显著敏感性。在已公开的 CSV 网格上，仅改变**NIST 网格指数单位**下的配对容差（0.0 与 15.0），即可在事件表不变的前提下使二元 CHSH 统计量出现约 +0.50 量级的位移。

1. Introduction: The 30cm Truth / 引言：30厘米的真相

Units (critical): This deposit pairs events using a tolerance measured on the CSV `t` grid — **NIST grid-index units**, not calibrated nanoseconds. The laboratory heuristic "30 cm truth" (about **1 ns ↔ ~30 cm** in vacuum) motivates why tiny *physical* timing rules move coincidence counts; it does **not** identify the parameter `15.0` with "15 ns" here.

单位（关键）： 本归档在 CSV 的 `t` 网格上按容差配对——使用**NIST 网格指数单位**，而非本包内标定的“纳秒”。实验室启发中的“30 厘米真相”（真空中约 **1 纳秒 ↔ 约 30 厘米**）用于说明为何极小的物理时间规则会搬动符合计数；它**不**等于声称此处的参数 `15.0` 就是“15 纳秒”。

2. Methodology: Parallel Protocol Audit / 方法论：并行协议审计

We processed the NIST complete-blind event stream using two parallel protocols on the identical event list: Strict (pairing tolerance **0.0**) and Standard (**15.0** in the **same grid-index units**). This isolates bookkeeping (which pairs enter the CHSH denominator) from any change to the underlying event generator.

我们在完全相同的事件表上运行两个并行协议：严苛（配对容差 **0.0**）与标准（**15.0**，**同一网格指数单位**）。这把“会计规则”（哪些配对进入 CHSH 分母）与事件生成机制本身的变化分离开来。

3. Results: Protocol-Induced Shift / 结果：协议引起的统计位移

On the frozen `battle_result.json` snapshot: **S_strict = 2.336276** (tolerance 0.0), **S_standard = 2.839387** (15.0), **Δ = +0.503111**. Accepted pair counts: strict **136632** vs standard **148670** — the wider tolerance changes the CHSH denominator, not the raw row list.

冻结 `battle_result.json`：**S_strict = 2.336276**（容差 0.0），**S_standard = 2.839387**（15.0），**Δ = +0.503111**。接受配对数：严苛 **136632**、标准 **148670**——更宽容差改变 CHSH 分母，而非原始行表。

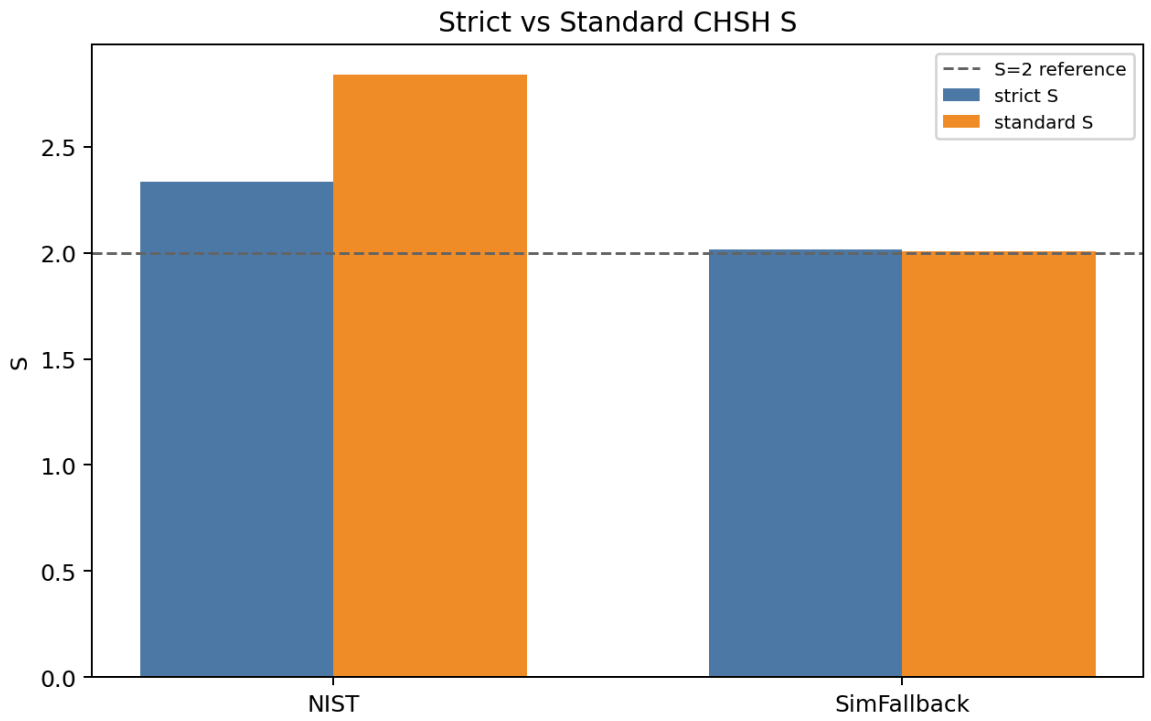

Figure 1: CHSH S under strict vs standard tolerances (grid-index units). / 图 1：严苛与标准容差（网格指数单位）下的 CHSH S。

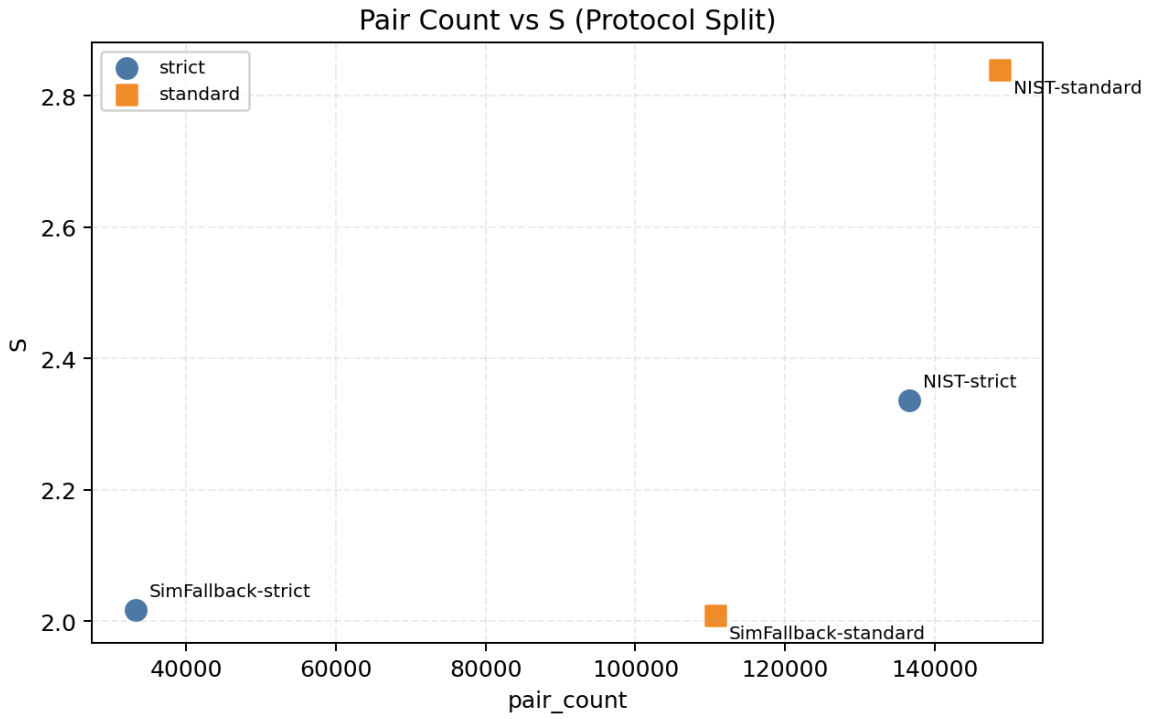

*Figure 2: Pair-count vs S for the two protocols (strict **136632**, standard **148670**). / 图 2：两协议配对数与 S（严苛 **136632**，标准 **148670**）。*

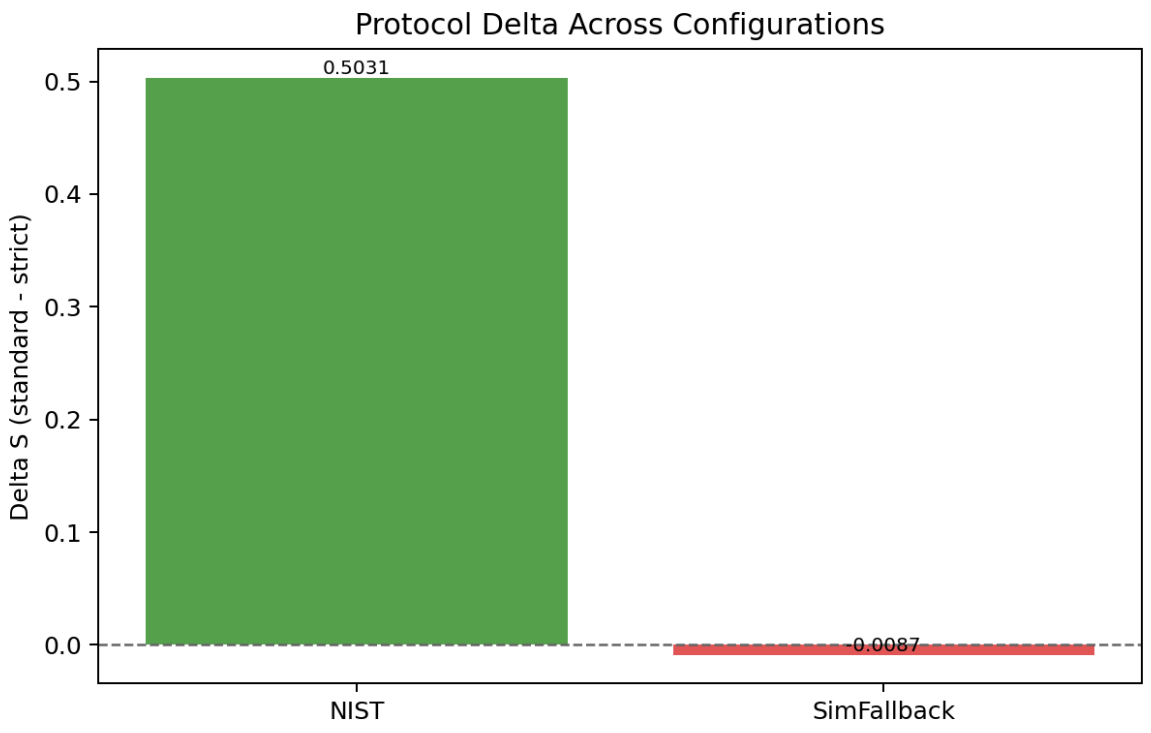

Figure 3: ΔS = +0.503111 (bookkeeping shift). Standard-branch bootstrap CIs include 2√2. / 图 3：ΔS = +0.503111（账目位移）。标准分支 bootstrap 区间覆盖 2√2。

> *"If the reported value changes when the ledger changes, protocol definitions must be audited alongside physical interpretation." / "若报告值会随记账规则变化，则在物理解释之外必须同时审计协议定义。"*

3.1 Tsirelson / bootstrap / Tsirelson 与 bootstrap

The point estimate `S_standard = 2.839387` slightly exceeds `2√2`, but `chsh_bootstrap_ci_standard15.json` has `ci_contains_tsirelson: true`. Read as **estimator + finite-sample sensitivity**, not a super-quantum device claim.

点估计略大于 `2√2`，但 `chsh_bootstrap_ci_standard15.json` 中 `ci_contains_tsirelson: true`。应理解为**估计量与有限样本敏感性**，非"超量子装置"主张。

4. Discussion: The Geometric Illusion (interpretive hypothesis) / 讨论：几何幻觉（解释性假说）

Interpretive hypothesis — not a finding of §2–§3. The audit establishes **protocol-dependent** *S* on one NIST stream. The paragraphs below

解释性假说——并非第2–3节的审计结论。上文仅在该 NIST 事件流上确立 *S* 的**协议依赖性**。下文是我们受该敏感性启发、但**并未由此证明的推测性**读法；作为**可证伪的**

are our **speculative** reading—motivated by that sensitivity, not proved by it. We state them as **falsifiable intuition** for follow-up; **we do not assert their physical truth here**. **Not operational conclusions**—hypotheses to test.

研究直觉提出，**本文不对其物理真实性作定论。非可执行结论**——待检验假说。

Hypothesis: Bell violation *as a headline metric* can behave like a geometric or bookkeeping illusion: binarizing continuous fields through time gates and pairing rules may sample pipeline geometry as much as—or more than—any intended non-local story. **This is suspicion, not a measurement conclusion from § 3.**

假说：作为头条指标的 Bell 违背*可能*呈现几何或账目式幻觉：通过时间门与配对规则对连续场二值化，采样的或许是流程几何，未必等同于意图中的非局域叙事。**这是怀疑，而非 § 3测量结论。**

Hypothesis: *If* that reading were broadly right, programs that lean on peak *S* after aggressive inclusion rules might be harnessing **bookkeeping-sensitive** statistics—*if* whole-sample fairness matters at scale, contrasts can shrink when denominators and discards are fully logged. **Conditional reasoning only, not an empirical prediction about any hardware line.**

假说：*若*该读法更广成立，则依赖激进纳入后峰值 *S* 的路线可能利用**对账目敏感**的统计量——*若*规模化需更公平计入全样本，分母与丢弃被完整记录时对比可能减弱。**仅为条件推理，非对任一硬件路线的经验预言。**

5. Conclusion: A Boundary-Aware Audit Guide / 结论：一份边界清晰的审计指南

This workflow shows the reported CHSH statistic on this NIST stream is a **protocol-dependent object**. If entanglement narratives lean on such metrics without denominator transparency, “quantum correlation” can track bookkeeping as much as mechanism.

本工作流表明：该 NIST 事件流上报告的 CHSH 统计量是**依赖于协议的对象**。若纠缠叙事仅依赖此类指标却缺乏分母透明度，“量子关联”可能与账目规则同步波动，而不仅是机制。

6. Reproducibility / 最小复现

Repository / 仓库： https://github.com/tomnattle/chain-explosion-model

EN Data: `data/nist_completeblind_side_streams.csv` • Run: `python scripts/explore/chsh_denominator_audit.py` • Out: `artifacts/bell_window_scan_v1/` • Frozen: `battle_results/nist_completeblind_2015-09-19/battle_result.json` • Bootstrap: `artifacts/reports/chsh_bootstrap_ci_strict0.json`, `chsh_bootstrap_ci_standard15.json` • Optional gate audit: `python scripts/explore/bell_honest_gate_audit_v1.py`

中 数据：`data/nist_completeblind_side_streams.csv` • 运行：`python scripts/explore/chsh_denominator_audit.py` • 输出：`artifacts/bell_window_scan_v1/` • 冻结：`battle_results/nist_completeblind_2015-09-19/battle_result.json` • Bootstrap：`artifacts/reports/chsh_bootstrap_ci_strict0.json`、`chsh_bootstrap_ci_standard15.json` • 可选门控审计：`python scripts/explore/bell_honest_gate_audit_v1.py`

Verification script / 验证脚本: `scripts/explore/chsh_denominator_audit.py`
Zenodo / DOI: 10.5281/zenodo.19784937 • record
GitHub / 仓库: https://github.com/tomnattle/chain-explosion-model